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(54) **CONTINUOUS DISPENSING REEL SYSTEM**

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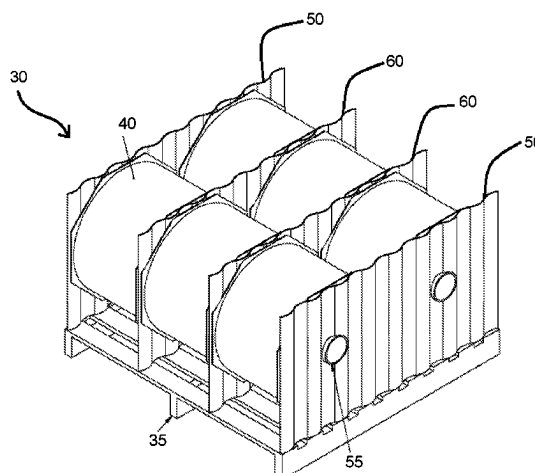
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(57) **ABSTRACT**

A generally continuous string of container carriers are
positioned on a base (35) wherein container carriers are
housed on multiple reels (40) and are spliced together
between adjacent reels to form the generally continuous
string and the multiple reels are arranged on at least one axle
(55) for dispensing.

21 Claims, 5 Drawing Sheets



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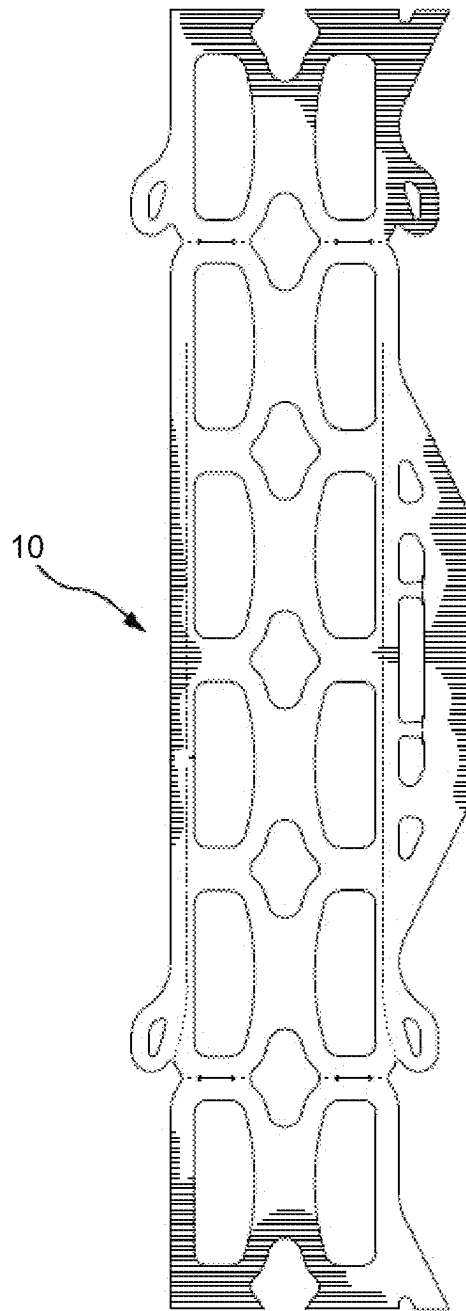


FIG. 1
(PRIOR ART)

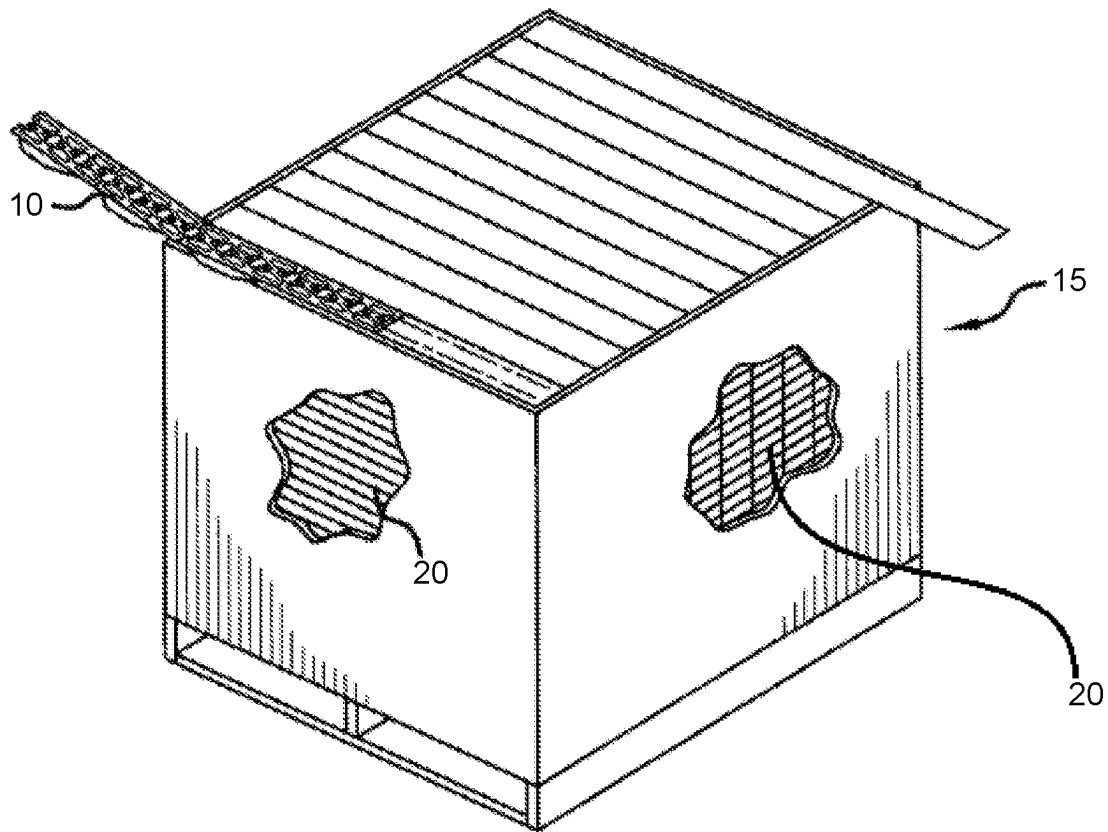


FIG. 2
(PRIOR ART)

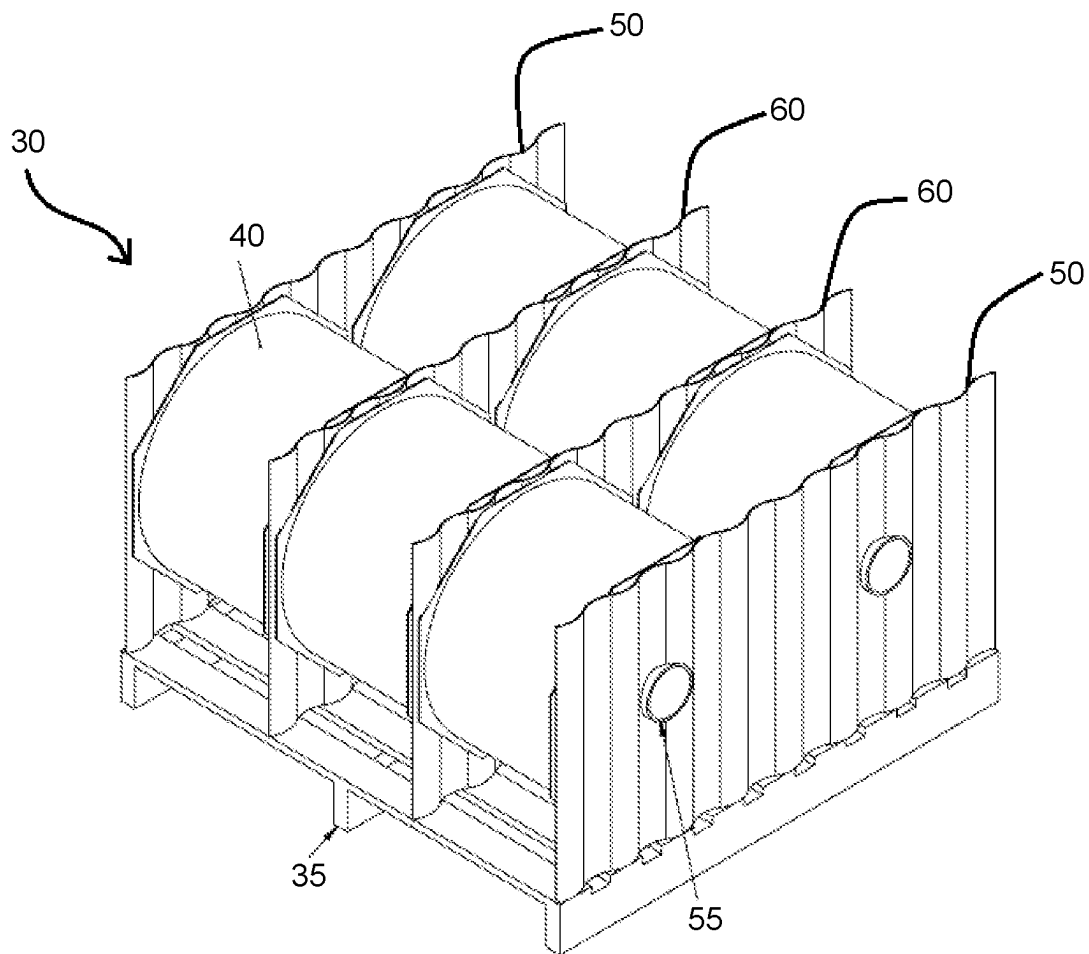


FIG. 3

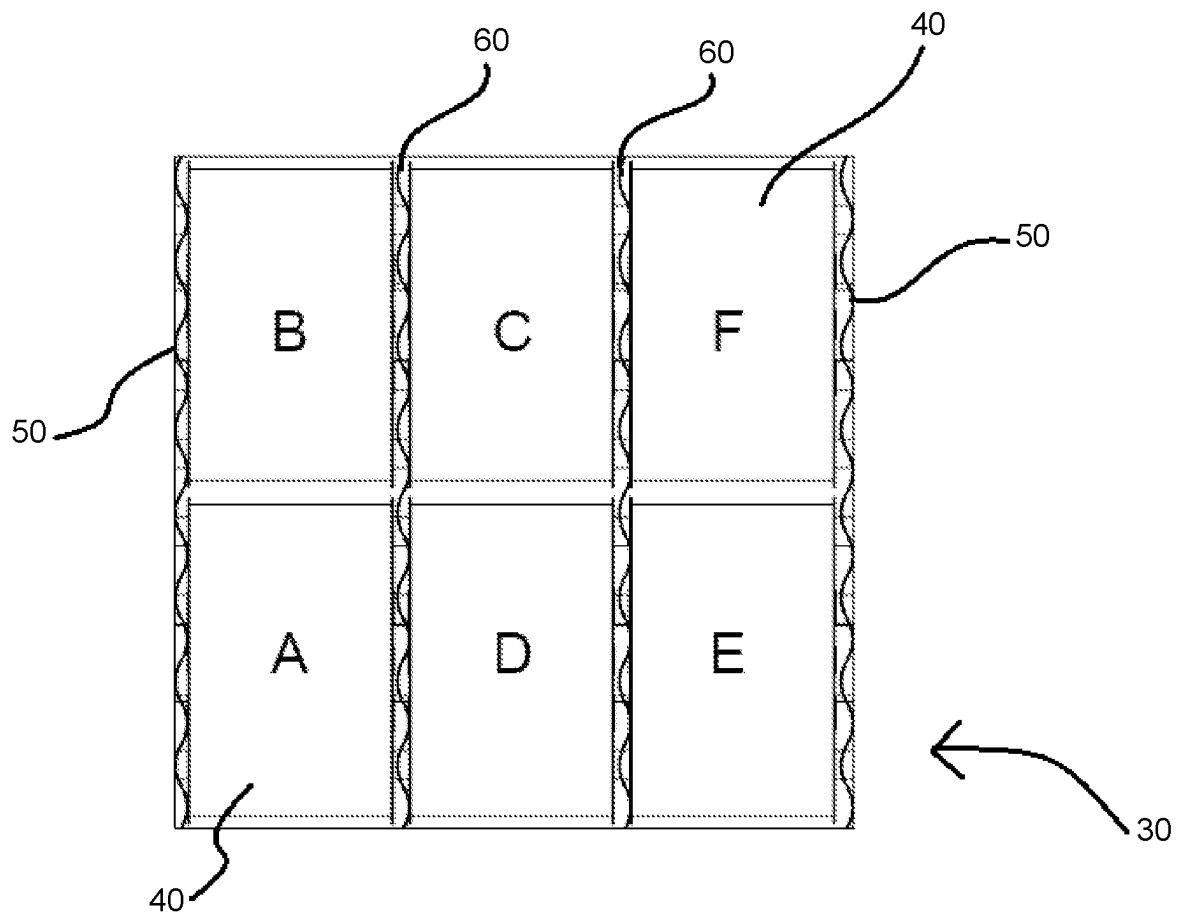


FIG. 4

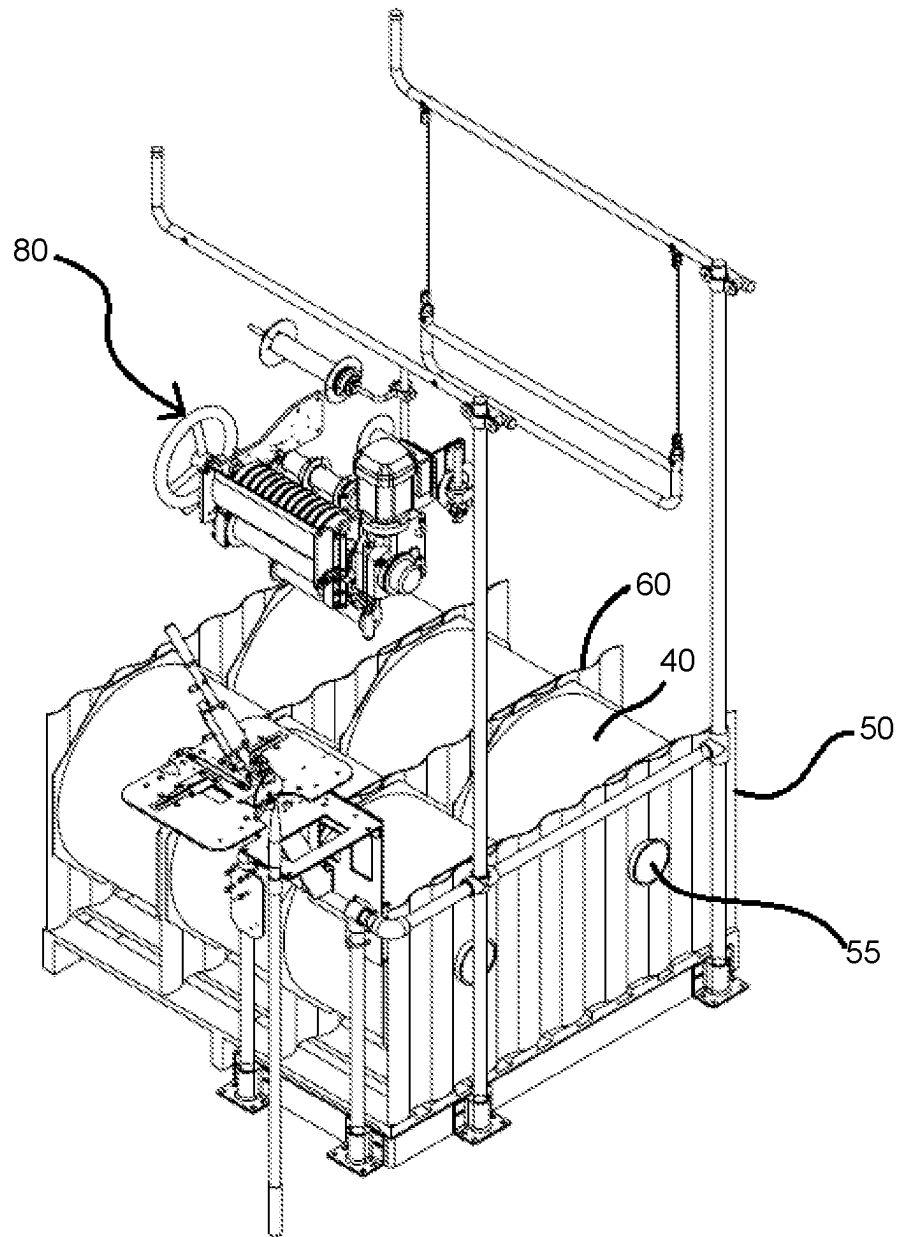


FIG. 5

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CONTINUOUS DISPENSING REEL SYSTEM**CROSS REFERENCE TO RELATED APPLICATION**

This application is a National Phase filing of PCT International Patent Application Number PCT/US2020/0551922, filed on 12 Oct. 2020, which claims the priority of U.S. Provisional Patent Application Ser. No. 62/947,822, filed 13 Dec. 2019.

These applications are hereby incorporated by reference herein in their entirety and is made a part hereof, including but not limited to those portions which specifically appear hereinafter.

BACKGROUND OF THE INVENTION**Field of the Invention**

This invention relates to an apparatus and method for dispensing container carriers in a space-efficient manner.

Description of Prior Art

Container carriers are used to unitize a plurality of containers. Typical containers are bottles, cans and other containers having a sidewall and a neck, chime or raised rib around an upper portion of the container. Container carriers typically connect two or more containers into a sturdy unitized package of containers. Container carriers are generally planar arrays of rings, sometimes referred to as “six-pack carriers,” and may be formed from a thermoplastic sheet material.

The thermoplastic sheet material is typically extruded and then punched so that large numbers of container carriers are formed end to end resulting in continuous elongated strings or strips of container carriers. Unless specified otherwise, container carriers as used in this specification are defined as the continuous elongated string of container carriers prior to application onto containers and subsequent division into individual container carriers.

Prior art methods of packaging container carriers involve accumulating the elongated strings of container carriers onto reels. The reels of container carriers are unwound at a later time during application onto containers. The reel method of storing and applying the container carriers to containers requires splicing the end of one reel with the beginning of the following reel without interrupting the application process. In addition, the reel method of storing and applying carriers is generally limited to reels of a size which may be physically lifted and manipulated by the applying machine operator. Also, the reel unwinding equipment must accommodate the inertia and resultant backlash inherent in unwinding a wound strip of material. Further, the generally circular or octagonal shape of the reels limits the number of reels which may be assembled onto a single pallet for shipment between the production facility and the application facility.

According to another prior art method of storing and dispensing container carriers, such as disclosed in Wanderer, U.S. Pat. No. 3,285,405, and Slaters Jr. et al., U.S. Pat. No. 6,068,125, incorporated by reference herein, elongated strings of container carriers are fan folded, similar to pin-feed computer paper, into boxes or cartons. The container carriers are dispensed from the boxes or cartons during the

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application process and successive boxes or cartons containing stacks of container carriers are spliced together as the boxes empty.

However, such methods for storing and dispensing container carriers may lead to uneven accumulation of the container carriers at each end of the box or carton, specifically where the container carriers are folded over onto each other. As a result, cartons may suffer from uneven stacks of folded container carriers whereby a central portion of the carton is not filled to capacity and end portions of the carton include distinctly higher stacks of carriers. This problem is referred to as “birdsnesting” and may result in inefficiently packed cartons of container carriers and even tangling of the string of container carriers during the unwinding process. In addition, the equipment required to properly fan fold and stack container carriers into cartons is sophisticated and requires substantial investment and maintenance for continuous operation.

SUMMARY OF THE INVENTION

It is one object of this invention to provide a package of container carriers which allows for space-efficient storage of reels of container carriers and permits uninterrupted removal of a mostly continuous elongated strip of container carriers.

A package of container carriers for shipment and storage and subsequent application to groups of containers is constructed to contain reels of container carriers. Elongated strips of container carriers are preferably rolled into multiple reels and positioned and spliced within cartons for shipment and subsequent dispensing in a generally continuous manner.

A resulting package of container carriers according to an embodiment of the invention preferably includes a carton and a generally continuous string of container carriers rolled onto a plurality of reels in a generally continuous fashion wherein adjacent reels are separated by a splice.

BRIEF DESCRIPTION OF THE DRAWINGS

The above-mentioned and other features and objects of this invention will be better understood from the following detailed description taken in conjunction with the drawings wherein:

FIG. 1 is a side elevational view of a string of container carriers according to the prior art;

FIG. 2 is a side perspective view of a carton or package of container carriers according to the prior art;

FIG. 3 is a side perspective view of a carton or package of container carriers according to one preferred embodiment of this invention;

FIG. 4 is a planogram of a carton or package of container carriers according to one preferred embodiment of this invention; and

FIG. 5 is a side perspective view of a carton or package of container carriers and a payoff system according to one preferred embodiment of this invention.

DESCRIPTION OF PREFERRED EMBODIMENTS

FIG. 1 shows a string of container carriers **10** as the term is used in the claimed invention. Container carriers **10** are preferably arranged in a generally continuous elongated string or strip having weakened areas between adjacent individual container carriers. Individual container carriers include two or more container receiving apertures **15** which

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are used to engage and unitize respective groups of containers into multipackages such as six packs, twelve packs, etc.

FIG. 2 shows a carton of fan folded stacks 20 of container carriers 10 according to the prior art. The elongated strip of container carriers 10 are preferably fan folded back and forth in alternating opposite directions resulting in horizontally extending rows of container carriers 10 forming vertically extending stacks 20. Fan folding container carriers 10, like pin-feed computer paper, preferably results in stacks 20 wherein a large amount of container carriers 10 are folded into a minimum amount of space. During the production process, container carriers 10 are preferably formed in the generally continuous, elongated string for shipment and storage in packaging, such as carton or package 15. After shipment to an application facility, such as a soft drink bottler, container carriers 10 are preferably removed from the packaging and applied, on an applying machine, to groups of containers, such as cans, during which application container carriers 10 are separated from the elongated strip and into individual container carrier packages.

In a preferred embodiment of the invention shown in FIG. 3, package 30 comprises a carton or at least two or more rigid sidewalls 50. The carton may contain two or more reels 40 of container carriers 10. A method for assembling package 30 of container carriers 10 is required that results in an efficiently filled package 30 and permits uninterrupted removal of a mostly continuous elongated strip of container carriers 10. In one preferred embodiment of this invention, package 30 is loaded with multiple reels 40 of container carriers 10. Preferably, though not necessarily, each reel 40 of container carriers 10 is separated and/or positioned using one or more dividers 60 to maintain position of and spacing between the various reels 40 of container carriers 10. This configuration helps prevent tangling and other problems encountered during loading and unloading of package 30.

A system for dispensing a generally continuous string 10 of container carriers includes a base 35. The base 35 may comprise a standard pallet or may be specifically adapted to the system herein.

At least two rigid sidewalls 50 preferably extend from the base 35. The rigid sidewalls 50 may comprise corrugated load bearing cardboard or similar construction. Additional sidewalls may extend between the two rigid sidewalls 50 that may be less structural but serve to protect the contents from contamination or intrusion. Alternatively, a carton may be formed of four similar sidewalls, each forming a structural rigid support for the system.

One or more axles 55 may extend between the rigid sidewalls 50. The axles 55 may comprise thick wall cardboard tubes or may comprise a more rigid and reusable solution including steel, plastic or other suitable material that can support the substantial weight of two or more reels 40 of container carrier.

At least two reels 40 are preferably positioned around each axle 55. The reels 40 as described each include a string 10 of container carriers. During production, the container carriers are preferably formed, such as through a punch press or rotary die, into generally continuous strings that are wound onto the reels 40 and then positioned within the subject system. Adjacent reels 40 are preferably are spliced together, so that a tail of reel A is connected to a head of reel B to form a generally continuous string 10 of container carriers.

In a preferred embodiment of this invention, each additional reel 40 of container carriers 10 that is added to package 30 is connected to the previous reel 40 of container carriers 10. Preferably, an end or tail of a container carrier

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at a bottom of a first reel 40 of container carriers 10 is connected with a head or beginning container carrier at a top of an additional reel 40 of container carriers 10. The end container carrier is preferably connected or spliced with the beginning container carrier using a weld attachment or a heat seal. A planogram showing a preferred arrangement of reels 40 is shown in FIG. 4. Adjacent reels 40 of container carriers 10 may also be connected using other methods known to those having skill in the art.

Specifically, the system preferably comprises an array of reels 40, each reel 40 of the array including a string of container carriers spliced to a string of container carriers of at least one adjacent reel 40, such as shown in FIGS. 3 and 4.

In this specific embodiment, six reels 40 may be arranged in two rows, wherein two reels 40 along one rigid sidewall 50 include a spliced connection relative to each other and two reels 40 along another rigid sidewall include a spliced connection relative to each other. Further, the two reels 40 in a center of the axle 55 include a spliced connection relative to each other. As shown in FIG. 4, a preferred arrangement includes connections among Reels A-B-C-D-E-F in that order head-to-tail. This arrangement encourages continuous runoff without tangling or interruption.

In a preferred embodiment, each reel 40 is preferably fully "back wrapped" to enable continuous head-tail splicing. Specifically, each reel 40 is manufactured so that a mid-point of the carrier length is folded and wound at a core of the reel 40 so each turn of the reel pays off the length going to the application machine and a length to be pulled to the machine after the first half of the reel is depleted.

The system may further include a divider 60 positioned between each adjacent reel 40. The divider 60 may be positioned specifically between each adjacent reel that shares a common axle 55 and may prevent binding between the reels as they are unwound. Like the sidewalls 50, the dividers 60 may also be corrugated cardboard and assist in the structural integrity of the system.

In one embodiment, an additional layer of reels 40 may be positioned on top of the plurality of reels 40. In this arrangement, at least one reel 40 of container carriers in the additional layer of reels is spliced to at least one reel 40 of container carriers in the plurality of reels below.

According to one embodiment shown in FIG. 5, a payoff system 80 may be integrated with the subject system to provide a powered pull system. The payoff system 80 according to this invention preferably pulls material from the reel 40 at a rate matched to a speed of the application machine. Such a powered feed system preferably acts as an isolation mechanism for the flow of carrier. As a result, the powered feed or pull system would keep upstream or downstream changes in tension from effecting proper application of the carrier to containers.

While in the foregoing specification this invention has been described in relation to certain preferred embodiments thereof, and many details have been set forth for purposes of illustration, it will be apparent to those skilled in the art that the apparatus is susceptible to additional embodiments and that certain of the details described herein can be varied considerably without departing from the basic principles of the invention.

The invention claimed is:

1. A system for dispensing a generally continuous string of container carriers, each container carrier for unitizing a plurality of containers, the system comprising:
 - a base;
 - two rigid sidewalls extending from the base;

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an axle extending between the rigid sidewalls; and
at least two reels positioned around the axle, each reel
including a string of container carriers that are spliced
together to form a generally continuous string of con-
tainer carriers, wherein each reel is back-wrapped.

2. The system of claim 1 wherein the at least two reels
comprise an array of reels, each reel of the array including
the string of container carriers spliced to another string of
container carriers of at least one adjacent reel of the array to
form the generally continuous string of container carriers.

3. The system of claim 1 comprising an additional two
axles, each axle supporting at least two reels.

4. The system of claim 1 comprising a divider positioned
between each adjacent reel of the at least two reels.

5. The system of claim 1 comprising a divider positioned
between each adjacent reel of the at least two reels that are
positioned around the axle and that share that axle.

6. The system of claim 1 wherein the rigid sidewalls form
a portion of a carton containing the generally continuous
string of container carriers.

7. The system of claim 1 further comprising at least
additional sidewalls forming a carton.

8. The system of claim 1 comprising six reels arranged in
two rows, wherein two reels of the six reels along one rigid
sidewall of the two rigid sidewalls include a spliced con-
nection and two reels of the six reels along the other rigid
sidewall of the two rigid sidewalls include a spliced con-
nection.

9. The system of claim 8 wherein two reels of the six reels
in a center of the axle include a spliced connection.

10. The system of claim 1 wherein the rigid sidewalls are
corrugated.

11. The system of claim 1 wherein the axle comprises a
cardboard tube.

12. The system of claim 1 wherein an additional layer of
reels is positioned on top of the at least two reels.

13. The system of claim 12 wherein at least one reel of
container carriers in the additional layer of reels is spliced to
at least one reel of container carriers in the at least two reels
below.

14. A generally continuous string of container carriers,
each container carrier for unitizing a plurality of containers,
the generally continuous string of container carriers com-
prising:

a first reel having a string of container carriers;

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a second reel having a string of container carriers, the
second reel connected to the first reel with a splice,
wherein the first reel and the second reel are back-
wrapped;

an axle supporting at least one of the first reel and the
second reel; and

at least two sidewalls enclosing the first reel and the
second reel.

15. A package of container carriers comprising:
a carton;

a generally continuous string of container carriers rolled
onto a plurality of reels in a generally continuous
fashion wherein adjacent reels are separated by a splice,
wherein an additional layer of reels is positioned on top
of the plurality of reels, and wherein at least one reel of
container carriers in the additional layer of reels is
spliced to at least one reel of container carriers in the
plurality of reels below.

16. The system of claim 14 wherein each reel is back-
wrapped.

17. The package of claim 15 further comprising a pair of
axles, each axle supporting two or more reels of container
carriers.

18. The package of claim 17 further comprising a divider
positioned between each adjacent reel on an each axle.

19. The package of claim 17 further comprising a pair of
sidewalls between which each axle extends.

20. The package of claim 15 wherein the plurality of reels
are arranged in a two by three array of reels within the
carton.

21. A system for dispensing a generally continuous string
of container carriers, each container carrier for unitizing a
plurality of containers, the system comprising:

a base;

two rigid sidewalls extending from the base;

at least one axle extending between the rigid sidewalls;
and

six reels arranged in two rows, wherein two reels of the
six reels along one rigid sidewall of the two rigid
sidewalls include a spliced connection and two reels of
the six reels along the other rigid sidewall of the two
rigid sidewalls include a spliced connection, wherein at
least two reels are positioned around the at least one
axle, each reel including a string of container carriers
that are spliced together to form a generally continuous
string of container carriers.

* * * * *